When Do Monte Carlo Mean Estimates and Confidence Intervals Fail Under Heavy Tails?

A controlled simulation of tail index, sign asymmetry, and median-of-means block choice

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Abstract

We conduct a computational analysis of the behavior of the simple mean, studentized confidence intervals and the median of means (MoM) estimator when applied to simulated data from a controlled signed Pareto family. Sign asymmetry was varied across simulations, while the absolute tail index was held constant. Simulations were conducted using two tail indices (1.5 and 2.5), three sign probabilities (0.5, 0.75, 1.0), two sample sizes (300 and 3,000) and three block counts (3, 9 and 25). Each simulation was replicated 1,200 times, which produced 14,400 individual samples and 43,200 pairs of estimator comparisons. Median and mean absolute error are reported along with paired median loss difference and percentile bootstrap intervals for the paired loss difference. Additionally, we report the proportions of paired wins and Wilson intervals for these paired wins, as well as the direction of losses (i.e., whether MoM outperformed the simple mean), bias, interval coverage and directional miss rate. Our results demonstrate that there is significant sensitivity to the distribution of the data (shape) and/or the method used to evaluate the estimators. These results do not represent a novel estimator or a theoretical finding.

1. Background and previous work

Given a sequence of i.i.d. random variables having finite first moment, the law of large numbers implies that the sample mean will converge to its population counterpart. Given that such sequences have finite second moments, then according to the Central Limit Theorem, the classical square root n normal approximation can be justified. However, when sequences have infinite second moments, convergence can still occur even though the usual normal approximation fails. Therefore, our focus extends beyond asking whether the sample mean converges; rather we seek to understand how well specific point estimates and interval estimates perform at realistic finite sample sizes.

In Logan et al. (1973) researchers investigated limiting distributions of self-normalized sums in heavy-tailed distributions demonstrating that Studentization cannot be assumed to produce normal approximations for all distributions. Lugosi & Mendelson (2019) provide an overview of methods to estimate robust means with particular emphasis on median of means methods. While these authors establish performance guarantees for finite variances, they emphasize that similar performance guarantees need not be transferable to infinite variances. Lecué & Lerasle (2020) investigate median-of-means based estimation methods for machine learning applications. This paper represents a controlled computational investigation within an existing body of literature; thus, we are not discovering some new phenomenon.

2. Distribution Family & Exact Targets

Consider a Pareto Type I distribution where $Y \sim \text{Pareto}(\alpha, \text{min} = 1)$, defined as $P(Y > y) = y^{-\alpha}$ for $y \geq 1$. Next, define $S = +1$ with probability p or $S = -1$ with probability $1 - p$, independently of Y. Then, let $X = SY$. The true population mean is given as $\mu = (2p - 1)\alpha/(\alpha - 1)$; however, μ is valid only when $\alpha > 1$. Further, $\text{E}[Y^2]$ is finite only when $\alpha > 2$. Thus, when $\alpha = 1.5$, the mean exists but the variance does not exist; whereas when $\alpha =$

2.5, both the mean and variance exist. The absolute-value law of X is identical across p at fixed α. Changing p changes tail balance and other aspects of the distribution, including the mean; this is a controlled tail-sign intervention, not a perfect isolation of skewness alone. Due to this property of $\alpha = 1.5$ and $\alpha = 2.5$ being in two distinct regimes regarding $E[X^2]$, any comparison of raw errors between these values of α changes the magnitude law and should not be viewed as a scale-adjusted efficiency ranking. Dependence among successive observations, volatility clustering, estimating parameters and using actual market data are not modeled.

3. Estimator and interval procedures

The common or ordinary mean is defined by $m = \Sigma X_i / n$. MoM divides the data into k consecutive blocks of varying size that differ from each other by no more than one observation. Then MoM calculates the mean of each of the k blocks, finds their median, and then returns it. Each of the k variants utilize the same information set and the same target μ. The paired loss function is $D = |\text{MoM} - \mu| - |m - \mu|$. Negative D favors MoM over m under an absolute loss function. In addition to the negative D we will report the median absolute error, the proportion of paired wins, and the average signed MoM error.

For the conventional two-sided interval for a single mean value, we have $m \pm t_{n-1,\ 0.975} \cdot s/\sqrt{n}$ where s represents the sample standard deviation. The proportion of independently generated intervals that contain the true μ is referred to as coverage. This interval is NOT a MoM confidence interval. The justification of the asymptotic normality / Student's t-distribution is not valid for $\alpha = 1.5$; therefore reporting coverage is merely descriptive and has no implications regarding whether or not the nominal coverage level is correct outside the simulated family.

4. Design of simulation experiments and reproducibility

We conducted a primary experiment using α values in {1.5, 2.5}, p values in {0.5, 0.75, 1}, n values in {300, 3000}, k values in {3, 9, 25}, and 1200 replicates per α-p-n cell. Magnitude was shared among all the p values for each replicate while signs were randomly generated separately. This was done intentionally, which means there is statistical dependence across p values; we will calculate uncertainty associated with each of the fixed conditions based on independent replicates and NOT because they are independent across p values. The three k choices represent different levels of sensitivity rather than an optimum selected based upon actual data.

Each of the samples of α-n-replicate data were generated with a deterministic NumPy SeedSequence. The Python script writes out replicate level and summary CSV files and checks row count, coverage/miss event partition, and performs an analytical Pareto mean check. An additional numerical audit below shows that the original controlled study CSV can be verified independently of the new data generation.

Although desirable prior to any external research submission, an independent software implementation would further strengthen our current analysis.

5. Results and Monte Carlo uncertainty

In Table 1 we report the median absolute loss difference (as well as a percentile-bootstrap 95% Monte Carlo interval (1,200 bootstrap resamples of independent replicates)) as well as the proportion of times MoM wins (along with a 95% Wilson binomial interval; Wilson, 1927). These summaries provide alternative measures to summarize paired loss differences which DO NOT require finite variance of paired loss differences. However, even though these summaries do not require finite variance of paired loss differences, bootstrap quantile inference (Efron, 1979) still requires independent replicates and sufficiently regular and locally continuous loss distributions; coverage is approximate at finite replication counts. We retain the sample-mean paired loss in the CSV only for descriptive purposes. We estimate coverage uncertainty as binomial across independent replicates for each fixed condition.

Table 1. Fixed $k = 9$, $n = 3{,}000$; 1,200 independent replications per condition. Paired D is the median of within-replication absolute-loss differences, not the difference of two medians.

α	p	Mean med.	MoM med.	Median paired D [95% MC]	Coverage	Below/above
1.5	0.50	0.125	0.102	-0.019 [-0.026, -0.012]	95.7%	2.4%/1.9%
1.5	0.75	0.132	0.121	-0.009 [-0.012, -0.002]	91.0%	8.5%/0.5%
1.5	1.00	0.165	0.205	+0.041 [+0.034, +0.049]	79.4%	20.6%/0.0%
2.5	0.50	0.027	0.032	+0.004 [+0.002, +0.006]	96.0%	1.9%/2.1%
2.5	0.75	0.024	0.029	+0.003 [+0.001, +0.004]	96.4%	2.0%/1.6%
2.5	1.00	0.016	0.020	+0.003 [+0.003, +0.004]	94.4%	4.9%/0.7%

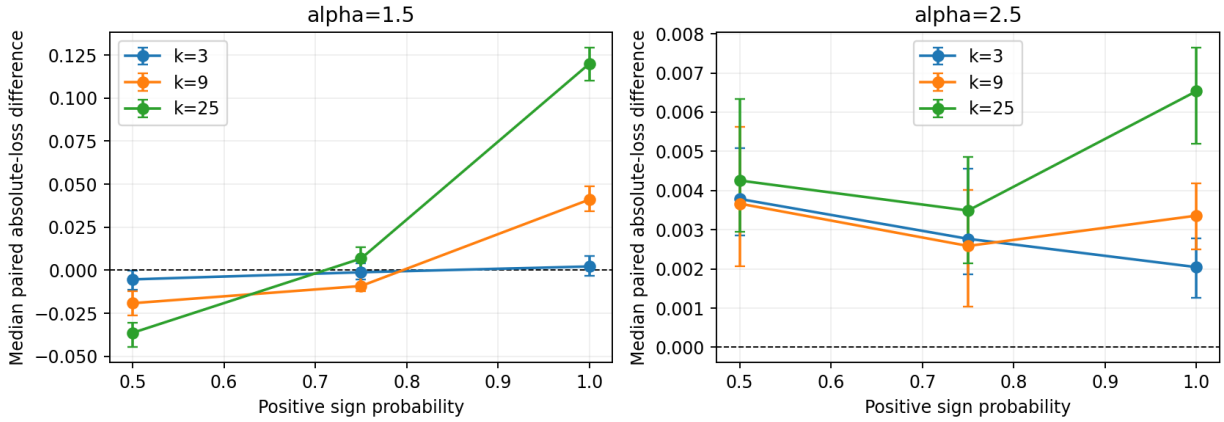


Figure 1. Median paired absolute-loss difference, with 95% percentile-bootstrap Monte Carlo intervals. Negative favors median-of-means.

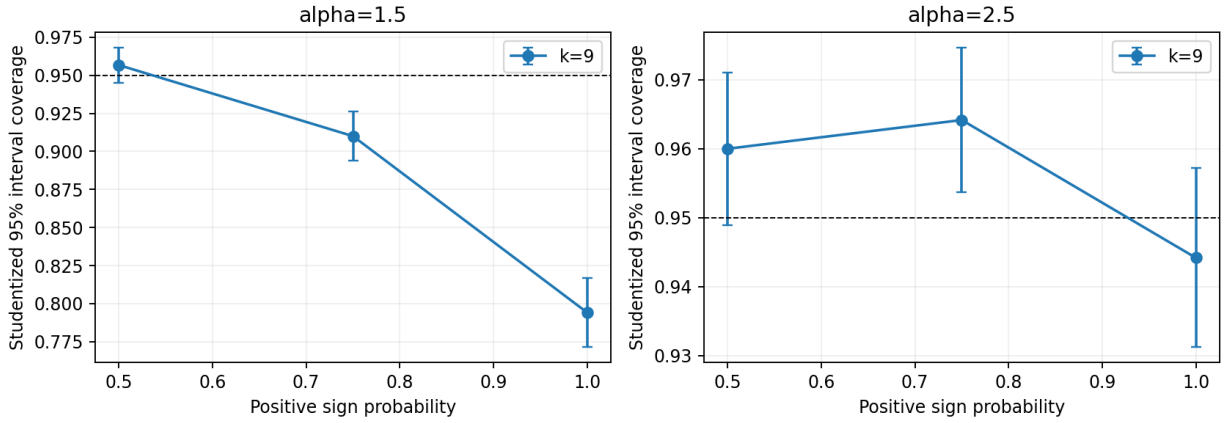


Figure 2. Empirical coverage with binomial Monte Carlo error bars; dotted line denotes nominal 95%.

6. Interpretation and sensitivity

The primary difference in interpretation with respect to the "typical" absolute error is the "expected" absolute loss. Because of the fact that rare extreme values are expected to contribute disproportionately to the mean but not the median, they may produce different outcomes. As well, MoM is known to create a trade-off between bias and robustness for distributions that are very highly skewed. In such cases, eliminating exceptionally high block averages will result in lower estimates compared to the true population mean.

The k-grid cannot be used to determine an optimal value of k without using an independent evaluation methodology. If the data follows a finite variance (e.g., $\alpha = 2.5$) then conventional methods for estimating means follow standard asymptotic theory; therefore, MoM typically provides less than maximum efficient estimates. However, when $\alpha = 1.5$, it is possible that MoM provides better empirical summary loss measures without inheriting finite-variance sub-Gaussian risk bounds (Devroye et al., 2016). Directional interval misses will help identify whether there is any degree of asymmetry present in the distribution but will not provide information about a particular cause of this asymmetry.

7. Verification, Limitations and Research Integrity

Deterministic regeneration was tested as well as testing against an exact population target. Coverage partitions were also tested and compared to prior studies. Additionally, the audit test provided as part of this package was run independently. The prior study involved 2,000 trials per cell while this consolidated study involves 1,200; the results from these two experiments should be considered as separate stochastic samples rather than combined into a single sample.

There are still several unresolved limitations including the signed-Pareto model being a stylized representation; finite number of Monte Carlo trials do not provide proof of asymptotic behavior; bootstrap intervals for the median of paired loss are approximate; the sign intervention affects more than just a single moment; normal approximation intervals for the mean of paired loss may fail to accurately estimate their coverage probabilities when sampling from an infinite variance distribution; no calibrated interval for MoM exists; and no demonstration of external predictive validity or trading applications exists. All of these limitations represent boundaries of scope rather than claims that could be addressed through additional text. This document represents an unpublished computational working paper.

8. Conclusion

In terms of controlled changes to tail index, tail-sign balance and block size, the reliability of estimated means and conventional confidence intervals depends upon the specific loss function and the assumed properties of the distribution. Median-of-means does not necessarily offer uniform superiority over conventional methods; in addition, studentized conventional confidence intervals can provide significantly poor coverage at finite sample sizes when sampling from infinite-variance distributions that are asymmetric. The contribution is a reproducible narrowly-scoped teaching/development experiment; not a new statistical method.

9. Scope and interpretation of generalizability

The signed Pareto intervention is a man made (synthetic) stress test. It is NOT a model of financial returns. Inferences in this study have been based upon independently drawn random values, known population averages and specific parametric families. There are several characteristics of financial return data that may limit applicability of the results herein. These include serial dependency, non-stationarity, time varying volatility and estimates of unknown parameters. Therefore, the study only supports statements regarding the methods employed in this study under the described synthetic conditions. For real return data, you will need to pose another research question, estimate uncertainty while accounting for dependencies in the data and use appropriate safeguards against post hoc selection bias.

A narrow, well-reproducible finding is better than an unsubstantiated claim of trading viability.

Additional qualifications to make additional inferences: Wilson intervals estimate the probability that the MoM has less absolute error than the ordinary average for some randomly sampled value(s) from the model used. They do NOT establish that the median estimator is optimal with respect to expected loss. Bootstraps provide intervals for the median of paired loss difference samples for multiple independent Monte Carlo experiments. They are NOT confidence intervals for the unknown population mean.

References

Devroye, L., Lerasle, M., Lugosi, G., & Oliveira, R. I. (2016). Sub-Gaussian mean estimators. *Annals of Statistics, 44*(6), 2695–2725. https://doi.org/10.1214/16-AOS1440

Efron, B. (1979). Bootstrap methods: Another look at the jackknife. *Annals of Statistics, 7*(1), 1–26. https://doi.org/10.1214/aos/1176344552

Lecué, G., & Lerasle, M. (2020). Robust machine learning by median-of-means: Theory and practice. *Annals of Statistics, 48*(2), 906–931. https://doi.org/10.1214/19-AOS1828

Logan, B. F., Mallows, C. L., Rice, S. O., & Shepp, L. A. (1973). Limit distributions of self-normalized sums. *Annals of Probability, 1*(5), 788–809. https://doi.org/10.1214/aop/1176996846

Lugosi, G., & Mendelson, S. (2019). Mean estimation and regression under heavy-tailed distributions: A survey. *Foundations of Computational Mathematics, 19*(5), 1145–1190. https://doi.org/10.1007/s10208-019-09427-x

Wilson, E. B. (1927). Probable inference, the law of succession, and statistical inference. *Journal of the American Statistical Association, 22*(158), 209–212. https://doi.org/10.1080/01621459.1927.10502953